ARNAB BASAK

Software Engineering Manager · Technical Project and Product Delivery

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Software engineering manager and delivery lead at a software services company, running end-to-end delivery for three external clients at the same time: a state-owned bank's digital banking platform (640,000+ retail customers, 500+ corporate customers, 1,500+ branch users), the national insurance agent licensing system for the regulator (800,000+ agents, 83 insurers), and a telecom interconnect billing platform. Lead a 16-person cross-functional team across backend, web, mobile, QA, production support and data, owning scope, timelines, resourcing, budget and the client relationship, and led the banking platform's UK expansion project with UK stakeholders.
Around seven years in software delivery, with nearly six on client projects under five regulators: central bank, tax, insurance, telecom and microcredit. Run Scrum with capacity-based sprint planning, a definition of ready, regular backlog refinement, risk and dependency tracking, and UAT and release governance in Jira and GitLab, and introduced an agentic SDLC with AI coding agents. Delivered results clients could measure: licence approval cut from 14 days to 24 hours, release time from one hour to 15 minutes, and 90% automated test coverage as a quality gate. Google Project Management Professional Certificate.

Work Experience

Celloscope Ltd Dhaka, Bangladesh

Senior Software Engineer (Software Engineering Manager, FinTech Portfolio) Jan 2024 – Present

- Delivery leadership: plan, execute and deliver three concurrent client projects end to end, from inception to closure, a digital banking platform for a state-owned bank, the national insurance agent licensing system for the regulator and a telecom billing platform, owning scope, timelines and resource plans.
- On time, within scope and on budget: Agent Licensing Online, the Customs Agent Management System, UniVerge and every Internet Banking feature shipped so far went live on schedule, within the agreed scope and within the defined budget.
- Scope, planning and documentation: define scope, milestones, deliverables and resource needs with clients and technical leads; manage a contracted scope of about 150 features with feature IDs, a definition of ready and T-shirt sizing; write and maintain PRDs and project documentation for every feature and product to agreed quality and compliance standards; plan sprints at an 80% velocity factor against roughly 85% realised capacity to protect committed dates.
- Agile execution and compliance: run Scrum across three release streams, covering sprint planning, stand-ups, reviews, retrospectives and a backlog refinement session every two sprints, and move every feature through the same stages: sprint, QA, UAT, production. Compliance requirements (Bangladesh Bank ICT security guidelines, applicable PCI-DSS controls, PII protection) are written into each PRD's definition of done and checked before every release.
- Risk, issue and dependency management: track risks, blockers and cross-team dependencies on the bank's core banking system, national payment networks (BEFTN, NPSB), digital wallets and government identity APIs, escalating early with options to keep integrations on schedule.
- Client and stakeholder management: main point of contact for a state-owned bank, the insurance regulator and a telecom operator; led the banking platform's UK expansion project, running meetings and requirement gathering with UK stakeholders to understand the market; sat with company management to gather requirements for the company's UK portfolio website, shaped it around the products the UK market needed, and designed it and led its development team; run status updates, demos and UAT sign-offs, and turn regulatory and business requirements into PRDs and acceptance criteria.
- Management reporting and communication: keep development teams, internal management and clients aligned through daily updates to stakeholders and the Managing Director on what happened, the impact, what is being done, who owns the next action and when it will be resolved; an automated business-status email on agreed metrics; and a weekly project and product status report to the Managing Director.
- Budget and resources: control headcount and software licences, and plan infrastructure capacity and cost, finalised in review with the relevant cross-functional teams.
- Quality, release and operations: 90% automated test coverage across backend, web and mobile, removing about 25 hours a week of manual QA; 15-minute standard releases instead of an hour, with downtime releases agreed with the client in advance; hands-on maintenance of the project's cloud infrastructure and VMs; ownership of SLAs, escalation and root-cause analysis with proactive monitoring.
- Team leadership and process: lead 16 direct reports across backend, web, mobile, QA, production support and data; hired 10+ engineers and own performance reviews and headcount. Introduced an agentic SDLC, using AI coding agents to simplify the team's process and my own PM work, with human review before anything is used.

Celloscope Ltd Dhaka, Bangladesh

Software Engineer Jan 2021 – Dec 2023

- Designed and developed the Bangladesh Customs Agent Management System with the team and led its delivery for the National Board of Revenue from day one to project closure and handover, covering 300k+ agents across 19 customs offices and cutting manual processing time by 90% through ASYCUDA World integration.
- Led and developed Agent Licensing Online with the team for the insurance regulator from day one to project closure and handover, digitising approvals for 800,000+ agents across 83 insurers and cutting licence turnaround from 14 days to 24 hours.
- Rebuilt the UniVerge ICX billing pipeline to process session detail records, deployed on high-availability infrastructure for 24/7 operation.
- Automated CI/CD workflows, reducing deployment time and enabling zero-downtime releases for 8+ projects.
- Mentored 6 engineers on coding standards and agile practices, improving code review efficiency by 40%.

Advanced Apps Bangladesh Ltd Dhaka, Bangladesh

Software Engineer Dec 2020 – Jan 2021

- Gathered requirements from a US-based client and developed FactFinder, a financial consultancy platform, with a Laravel/MySQL backend; also built cross-platform Flutter applications for logistics and retail clients.

Spectrum Engineering Consortium Ltd Dhaka, Bangladesh

Junior Software Programmer Jan 2020 – Nov 2020

- For the City of Cambridge, Ontario (Canada): built a malware scanner for its web portal that blocked 1.5k+ malicious uploads in 6 months, and developed the Service Cambridge mobile app (rated 4.6/5).
- Automated CI/CD pipelines, slashing deployment time from 1 hour to 15 minutes for 4+ projects.
- Engineered a Python/Shell SaaS platform for Bangladesh's National Board of Revenue, automating communication across 5+ legacy systems and reducing manual monitoring and errors by 90%.

Education

Master of Science (MSc.) in Computer Science (CS) Apr 2021 – Jul 2022

Jahangirnagar University Dhaka, Bangladesh

Bachelor of Science (BSc.) in Computer Science & Engineering (CSE) May 2016 – Apr 2020

American International University-Bangladesh (AIUB) Dhaka, Bangladesh

Skills

Project Management: Agile/Scrum, backlog refinement, scope, schedule and budget management, resource and capacity planning, risk, issue and dependency management, stakeholder management and reporting, PRDs and project documentation, UAT and release management, project closure and handover; Jira and GitLab for project tracking, AI coding agents (agentic SDLC)

Frameworks: Spring Boot, Node.js, Express.js, NestJS, FastAPI, Flutter

Integration: REST, SOAP, OAuth2, JWT, RBAC, Biometric, Core Banking, Bangla QR, MFS, BEFTN, NPSB, Payment gateways, n8n workflow automation

Database: PostgreSQL, Oracle, MySQL, Redis, MongoDB, PL/SQL, Supabase

Infrastructure and Observability: cloud and on-premises VMs (Proxmox), Docker, GitLab CI/CD, Ansible, Nginx, Prometheus, Grafana, Graylog, Zipkin, Alertmanager, Zabbix

Compliance: Bangladesh Bank ICT security guidelines, PCI-DSS, PII protection, GDPR (groundwork for the UK expansion)

Programming Language: Java, Python, JavaScript/TypeScript, Bash, SQL

Career Key Projects

Internet Banking Solution – Agrani Bank PLC. | Java, Spring Boot, Angular, Flutter, PostgreSQL, Redis, Docker, Prometheus/Grafana/Graylog

- Delivery: lead a 16-person team delivering a contracted scope of about 150 features across retail, corporate, branch and education banking, each through sprint, QA, UAT and production.
- Scope growth: extended the platform from retail into corporate banking for 500+ institutional customers (role hierarchies, bulk disbursement, group account aggregation) and an education fee collection product for 7 institutions.
- Integration delivery: led the Bangla QR integration with Bangladesh Bank's national QR scheme (QR-to-QR transfer, QR cash withdrawal) and coordinated dependencies with the Core Banking System, bKash, Nagad, BEFTN, NPSB, SSLCommerz, ShurjoPay and the Election Commission.
- Customer and operations outcomes: 17,000+ eKYC registrations a month, biometric and T-PIN transaction authorisation, dispute management, and a web reconciliation portal that replaced terminal-based back-office work.
- Standing Instructions: led requirements, design and on-time delivery for every customer of the bank across all branches, not only internet banking users, automating recurring deductions such as scheme instalments and fixed-amount transfers.
- Quality and release: GitLab pipelines with automated health checks and rollback, SonarQube quality gates, and Cypress and Appium test automation.
- Observability: set up Prometheus, Grafana and Alertmanager hands-on to monitor the platform and alert the team immediately on unusual behaviour, so issues are identified and fixed before customers encounter them.
- Database performance and security: tuned the PostgreSQL configuration and every bottlenecked query to make the platform faster and more reliable, and hardened PostgreSQL's security parameters.

UniVerge ICX Billing Solutions | Java, River Framework, Node.js, Oracle, Angular, Zookeeper, Jolokia, Docker, CI/CD

- Delivery: led the rebuild of the call-record pipeline on the legacy Java River framework and a new Angular and Node.js web application, now run 24/7 under an annual maintenance contract with a team I lead.
- Scale: parses 985 files a cycle with 30,000+ session records each (about 3.5 GB a day) on high-availability infrastructure I specified: 96 cores, 256 GB RAM and 18 TB SSD.
- Scope: three invoice types with an approval step, re-rating, regulator reports and live operational controls, serving two interconnect operators from one codebase.
- Operations: a monitoring portal for real-time log analysis and service control, and a watchdog daemon that removed downtime during daily log processing.

Agent Licensing Online (ALO) – Insurance Development and Regulatory Authority (IDRA) | Node.js, Angular, PostgreSQL, PL/SQL, Python, Docker, CI/CD

- Delivery: owned requirements, architecture and delivery for the insurance regulator from first build in 2021 through project closure and handover in June 2026, turning the 2021 agent regulations into requirements and acceptance criteria and rolling out across 83 insurers and the training institutes.

- Outcome: digitised licensing for 800,000+ insurance agents and cut licence approval from 14 days to 24 hours; won the Best Innovation Award.
- Scope: recruitment, training, licensing and renewal with a three-step review at the regulator, Election Commission NID verification, SMS alerts, and role-based access for regulator staff, insurers, training institutes and agents.
- Delivery practice: GitLab CI/CD with manual promotion gates, rolling updates and automatic rollback, plus Prometheus, Grafana and Loki monitoring with business-level alerts to the team.

Certifications and Courses

- Google Project Management: Professional Certificate | Coursera
- Agile Project Management: Scrum Step by Step with Examples | Udemy
- Java Programming: Complete Beginner to Advanced | Udemy
- Complete Python Bootcamp: Go from zero to hero in Python 3 | Udemy
- The Complete Flutter Development Bootcamp with Dart | Udemy
- Cisco CCNA Certification Course | American International University-Bangladesh (AIUB)

Publication

An Augmented Reality (AR) based interactive learning media platform for kids Mar 2021

Conference: The 4th International Conference, Panyapiwat Institute of Management (PIM), Thailand

This study aims to introduce innovative, interactive learning methods for children using augmented reality (AR) as an alternative to traditional textbooks, while also familiarizing them with AR technology. The Implemented AR-based learning platform utilizes image marker processing for interactive learning, presenting information visually and audibly, with hand movement interaction. It is built using Unity, C#, AR, Vuforia Engine, and the Android SDK.

Honors and Awards

- 1st Position, Networking Contest, Computer Science (CS) Fest 2018, American International University-Bangladesh (AIUB)
- Awarded Top 15, Business Case Competition, Blockchain Olympiad Bangladesh (BCOLBD) 2020